CrowdWisdomTrading — X Sentiment Price Prediction Agent: Evaluation Report

Author: Jash Bharat Pasad (IIT Gandhinagar). Asset: Bitcoin (BTC/USDT, hourly). Forecast horizons n+0 … n+5 (1–6 hours ahead). Inputs: X sentiment and price only. All metrics below are out-of-sample from a purged expanding walk-forward; hyper-parameters were fixed before testing.

1. Executive summary

- **Data integrity first.** One of the suggested datasets ("Bitcoin Tweets 2025–2026", Kaggle) is the 2021–23 corpus with every timestamp shifted +1,826 days (10,593/10,774 sampled rows identical; tweets stamped Feb 2026 quote BTC at ~$45k vs a real ~$68k). It was rejected; the original correctly-dated 4.69M-tweet corpus was used.
- **Domain sentiment model.** Trained on 510k author-labelled StockTwits crypto posts: bull-vs-bear AUC 0.862 vs VADER 0.575; 3-class macro-F1 0.52 vs 0.35; ~20,441 posts/s on one CPU.
- **Direction: no edge.** Across 2,807 out-of-sample hours (2021-07-29 → 2023-01-10) no model beats a random walk (OOS $R^2 < 0$ at every horizon; directional accuracy 48–52%). Adding X sentiment to a price model does not significantly change accuracy (Diebold–Mariano $p > 0.18$).
- **Trading.** Every long/short variant loses after 5 bp costs. Best: full_lgbm|6h_band with Sharpe -0.67 (90% CI -2.28 to 0.86), max drawdown -32%, Deflated Sharpe 0.06; BTC buy-and-hold over the same 2021-23 window: Sharpe -0.91, max drawdown -77%.
- **Where X sentiment does help: risk.** Adding X attention/disagreement to the HAR volatility model lowers QLIKE from 0.667 to 0.621 (DM t = -3.54, p = 0.0002). Vol-adaptive intervals fix the under-coverage of constant-width intervals in calm hours (76% → 83% vs 80% target).
- **X follows price.** Sentiment correlates 0.22 with the previous hour's return but only ~0.04 with the next hour's (lead–lag chart). The raw 'high bearishness → negative returns' pattern is a regime artefact: demeaned within each week it reverses into a mild contrarian bounce.

2. Data

Source	What	Coverage / volume	Notes
Kaggle kaushiksuresh147 Bitcoin_tweets.csv	historical X posts (train/validate)	4.69M posts → 4,039 covered hours, 4,007 model-eligible	malformed CSV rows dropped: 4,487
Apify (live, last 180 days)	X posts, engagement, author followers	5,807 posts in 478 hourly samples, 2026-10-03 → 2026-10-31; cost $3.45	kaitoeasyapi actor (apidojo capped on free plan); budgeted hourly sampling
Binance spot archive (data.binance.vision)	BTCUSDT 1h OHLC	2021-01 → 2026-10	keyless, immutable files
Tavily (news)	recent BTC news	last 7 days	outlook only — not a model feature

Coverage gaps in the historical corpus (220 well-covered days; longest unbroken run 15 days) mean post counts reflect the collector's effort. Attention is therefore measured relative to a trailing covered-hour baseline, and a row is modelled only if the previous hour has ≥ 50 posts. 47% of posts are automation-like (non-human client, templated text, account < 30 days, hashtag stuffing, price-ticker templates).

3. Method (summary — full detail in docs/METHODOLOGY.md)

- Timing contract: decision at the open of hour n with posts before it and closes to n−1; target $y_h = \log(\text{close}_{n+h}/\text{close}_{n-1})$, so n+0 is a genuine 1-hour-ahead forecast. A unit test shocks all future data and asserts features are unchanged.
- Features (sentiment & price only): lagged returns, realised and Parkinson vol, drawdown; human-filtered net sentiment, bullishness index, bearish share (kept separate — negative-tone asymmetry), disagreement, bot share, abnormal attention, sentiment momentum/surprise, sentiment–price divergence.
- Model: one LightGBM per horizon on volatility-normalised targets (regime-comparable), Huber loss, heavy regularisation; split-conformal 80% intervals. Benchmarks on identical hours: random walk, price-only, sentiment-only, no-bot-filter, linear Ridge; HAR-RV for volatility.
- Validation: purged expanding walk-forward (initial 1,200 eligible hours, refit every 250, 6-hour purge). Deflated Sharpe uses all 12 directional strategy variants tried.

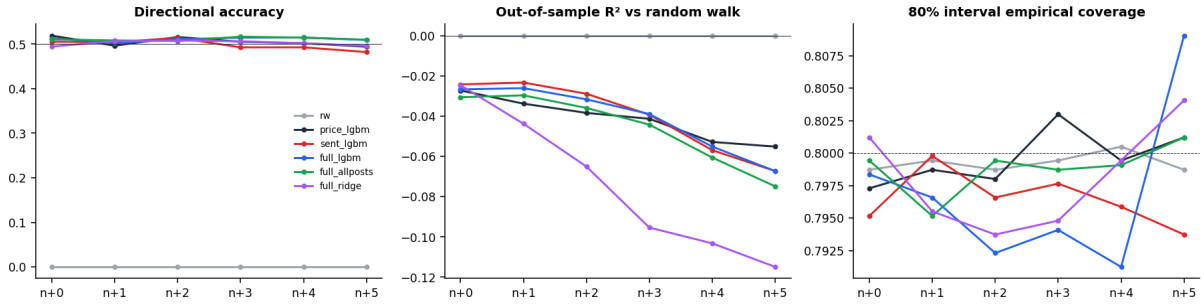
4. Forecast accuracy by horizon (out-of-sample)

Model	h	OOS R² vs RW	Dir. acc.	IC	Price MAPE %	PI80 cover	PI80 width %
rw	n+0	+0.0000	0.000	–	0.489	0.80	1.47
rw	n+2	+0.0000	0.000	–	0.860	0.80	2.73
rw	n+5	+0.0000	0.000	–	1.216	0.80	4.02
price_lgbm	n+0	-0.0272	0.519	+0.011	0.493	0.80	1.49
price_lgbm	n+2	-0.0384	0.516	+0.020	0.881	0.80	2.80
price_lgbm	n+5	-0.0551	0.494	+0.004	1.273	0.80	4.21
sent_lgbm	n+0	-0.0242	0.505	-0.005	0.495	0.80	1.50
sent_lgbm	n+2	-0.0289	0.514	+0.021	0.882	0.80	2.78
sent_lgbm	n+5	-0.0673	0.482	-0.042	1.270	0.79	4.08
full_lgbm	n+0	-0.0267	0.513	+0.006	0.495	0.80	1.50
full_lgbm	n+2	-0.0316	0.511	+0.022	0.888	0.79	2.77
full_lgbm	n+5	-0.0673	0.509	+0.005	1.283	0.81	4.22
full_allposts	n+0	-0.0306	0.511	+0.013	0.494	0.80	1.49
full_allposts	n+2	-0.0359	0.507	+0.024	0.890	0.80	2.82
full_allposts	n+5	-0.0749	0.510	-0.002	1.295	0.80	4.21
full_ridge	n+0	-0.0247	0.494	+0.001	0.499	0.80	1.51
full_ridge	n+2	-0.0650	0.508	+0.011	0.901	0.79	2.83
full_ridge	n+5	-0.1149	0.496	-0.012	1.315	0.80	4.32

Price MAPE is shown for completeness: it is tiny for every model (incl. the random walk) because P(t+h) ≈ P(t); it is not evidence of skill. OOS R² of returns vs a random walk is the meaningful number.

full_lgbm vs	n+0 (t, p)	n+2	n+5
rw	+3.27, 0.999	+2.24, 0.988	+3.04, 0.999
price_lgbm	-0.11, 0.456	-0.52, 0.300	+0.58, 0.721
full_allposts	-0.93, 0.177	-0.73, 0.233	-0.88, 0.191
full_ridge	+0.21, 0.585	-1.40, 0.081	-1.36, 0.087

Diebold–Mariano (squared error, HAC, HLN-corrected). Negative t ⇒ full_lgbm more accurate; p is one-sided.



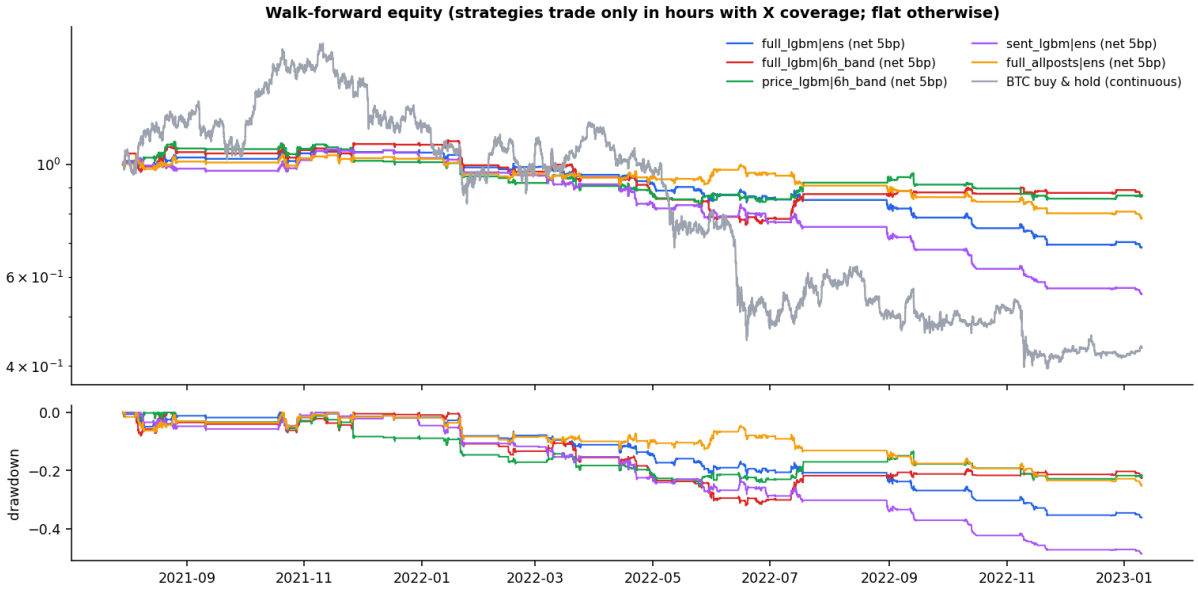
Directional accuracy, OOS R² and interval coverage by horizon.

5. Trading performance (required metrics)

Strategy	Sharpe	Sharpe 90% CI	Sortino	Max DD	Calmar	Turnover/day	DSR
price_lgbm|h0	-2.24	-3.74 … -0.9	-2.93	-43.8%	-0.76	1.68	0.000
price_lgbm|ens	-2.12	-3.49 … -0.7	-2.80	-42.7%	-0.72	1.16	0.000
sent_lgbm|h0	-3.20	-4.47 … -1.99	-4.15	-51.1%	-0.92	2.06	0.000
sent_lgbm|ens	-2.78	-4.23 … -1.44	-3.64	-48.6%	-0.84	1.76	0.000
full_lgbm|h0	-2.64	-3.86 … -1.46	-3.48	-44.1%	-0.86	1.87	0.000
full_lgbm|ens	-1.83	-3.06 … -0.58	-2.50	-36.2%	-0.72	1.30	0.001
full_allposts|h0	-1.99	-3.21 … -0.81	-2.76	-36.1%	-0.78	1.84	0.001
full_allposts|ens	-1.19	-2.43 … 0.02	-1.66	-25.3%	-0.67	1.26	0.013
full_ridge|h0	-1.13	-2.43 … 0.23	-1.60	-27.6%	-0.58	1.37	0.017

Strategy	Sharpe	Sharpe 90% CI	Sortino	Max DD	Calmar	Turnover/day	DSR
full_ridge|ens	-1.34	-2.65 … 0.02	-1.86	-30.8%	-0.63	1.10	0.008
price_lgbm|6h_band	-0.72	-2.09 … 0.6	-0.99	-25.3%	-0.39	0.36	0.051
full_lgbm|6h_band	-0.67	-2.28 … 0.86	-0.94	-32.0%	-0.31	0.37	0.056
buy_hold_covered_hours	-1.02	-2.43 … 0.33	-1.39	-55.0%	-0.61	0.32	–
buy_hold_continuous	-0.91	-2.24 … 0.39	-1.26	-77.2%	-0.74	0.00	–
full_lgbm|ens@0bps	-0.16	-1.4 … 1.12	-0.22	-17.1%	-0.13	1.30	–
full_lgbm|ens@10bps	-3.48	-4.67 … -2.21	-4.67	-53.2%	-0.94	1.30	–

Hourly net returns after 5 bp per unit turnover (rows marked @0bps/@10bps show cost sensitivity), annualised with √8760. Strategies are flat in hours without X coverage. h0 = 1h-ahead signal; ens = horizon-averaged signal; 6h_band = decide every 6h on the n+5 forecast with a 0.25 no-trade band (added after round 1, counted in DSR).



Walk-forward equity and drawdown.

6. Volatility: where X sentiment adds information

Model	MSE log-RV	QLIKE	R^2 vs HAR	corr
har	1.167	0.667	+0.000	0.636
har_x	1.177	0.621	-0.009	0.644
price_gbm	1.320	0.687	-0.132	0.592
full_gbm	1.345	0.658	-0.153	0.599

Comparison	DM on MSE (t, p)	DM on QLIKE (t, p)
har_x_vs_har	+0.78, 0.781	-3.54, 0.0002
full_gbm_vs_price_gbm	+1.20, 0.884	-1.73, 0.0416
full_gbm_vs_har	+2.54, 0.994	-0.24, 0.4036
price_gbm_vs_har	+2.33, 0.990	+0.49, 0.6876

Target: log realised variance over n..n+5. QLIKE (Patton 2011) is the robust loss for variance forecasts and penalises under-predicting volatility spikes. The gain is on QLIKE, not on MSE of log-RV.

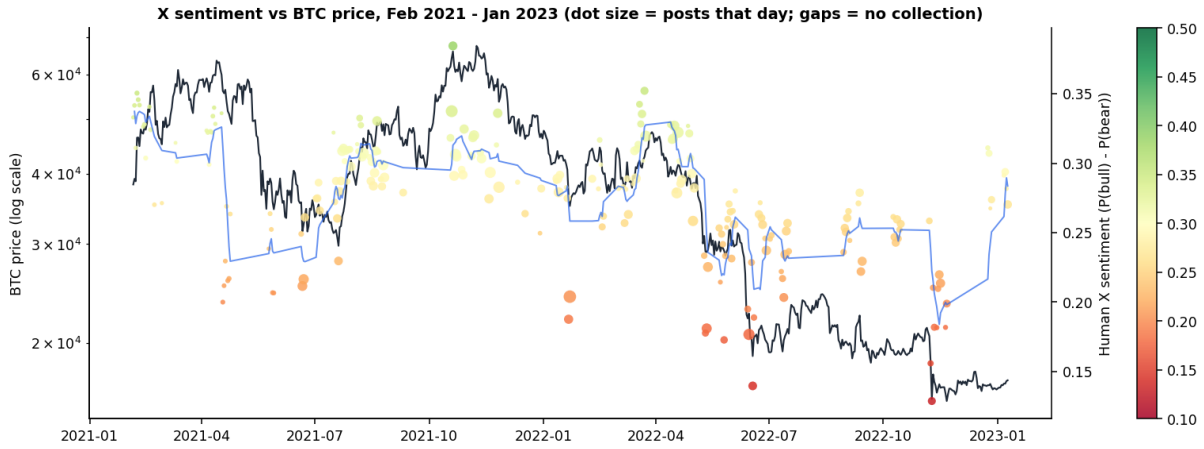
n+5 80% interval	coverage all	calm hours	turbulent hours	mean width %
adaptive	0.840	0.826	0.855	4.17
constant	0.810	0.758	0.868	4.03

Vol-managed long (covered hours)	Sharpe*	Sortino*	Max DD	Ann. vol*
long_unmanaged	-2.17	-2.95	-55.0%	0.70
vol_managed_har	-2.75	-3.65	-52.5%	0.62
vol_managed_har_x	-2.95	-3.95	-52.9%	0.60

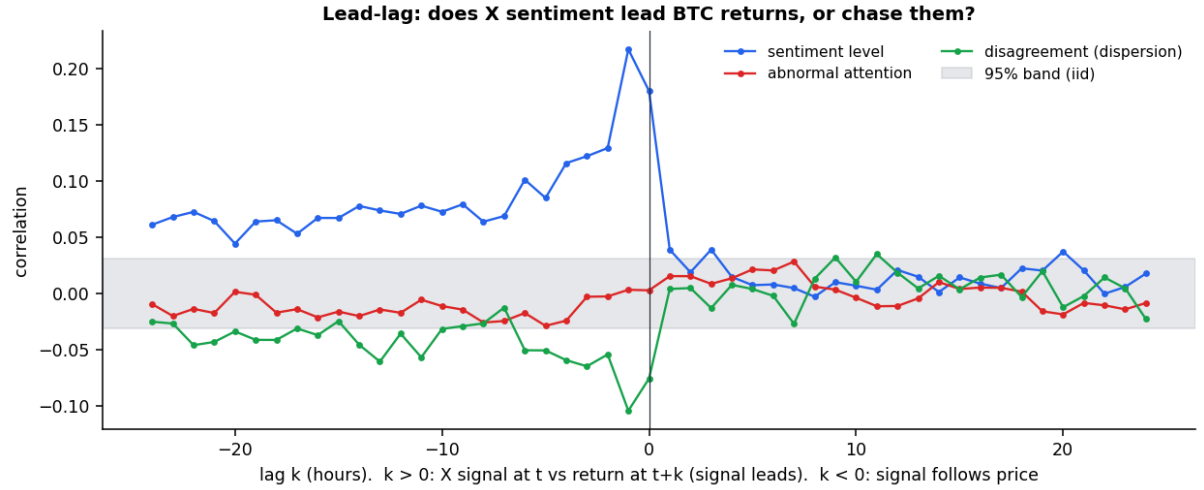
Vol-managed long (covered hours)	Sharpe*	Sortino*	Max DD	Ann. vol*
vol_managed_full_gbm	-3.92	-5.16	-58.9%	0.57

*annualised per active hour (covered hours only), so comparable across rows but not with §5. Better variance forecasts did not translate into better risk-adjusted returns in this mostly-bear sample: honest negative result.

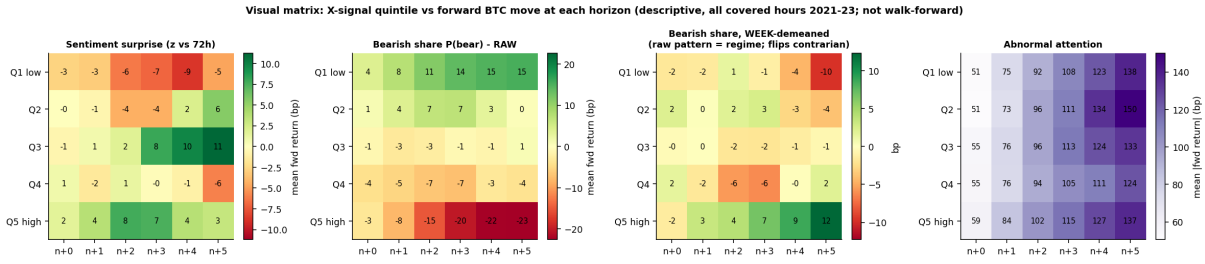
7. Visual matrix: X sentiment vs price through time



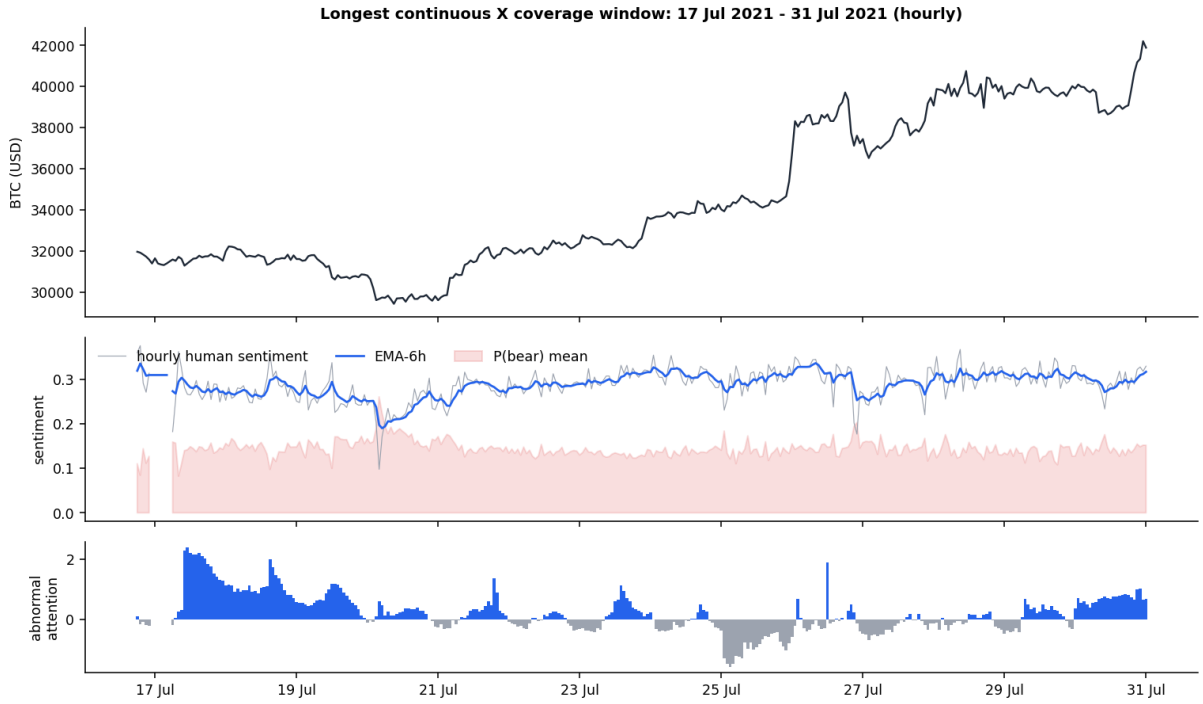
Daily human X sentiment vs BTC, 2021–23.



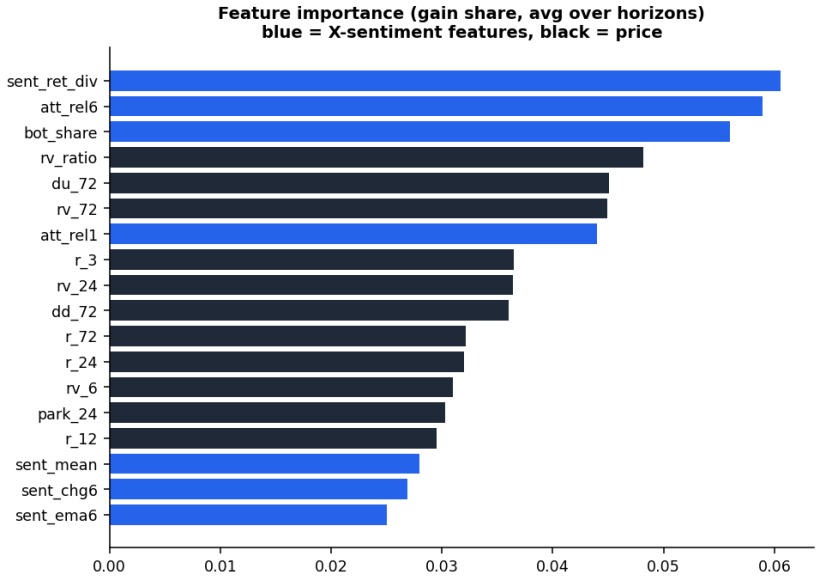
Lead–lag: sentiment reacts to past returns; almost no lead on future returns.



Signal quintile × horizon. Raw bearish-share pattern is a regime effect; week-demeaned it flips contrarian.



Hourly zoom on the longest continuous coverage window.

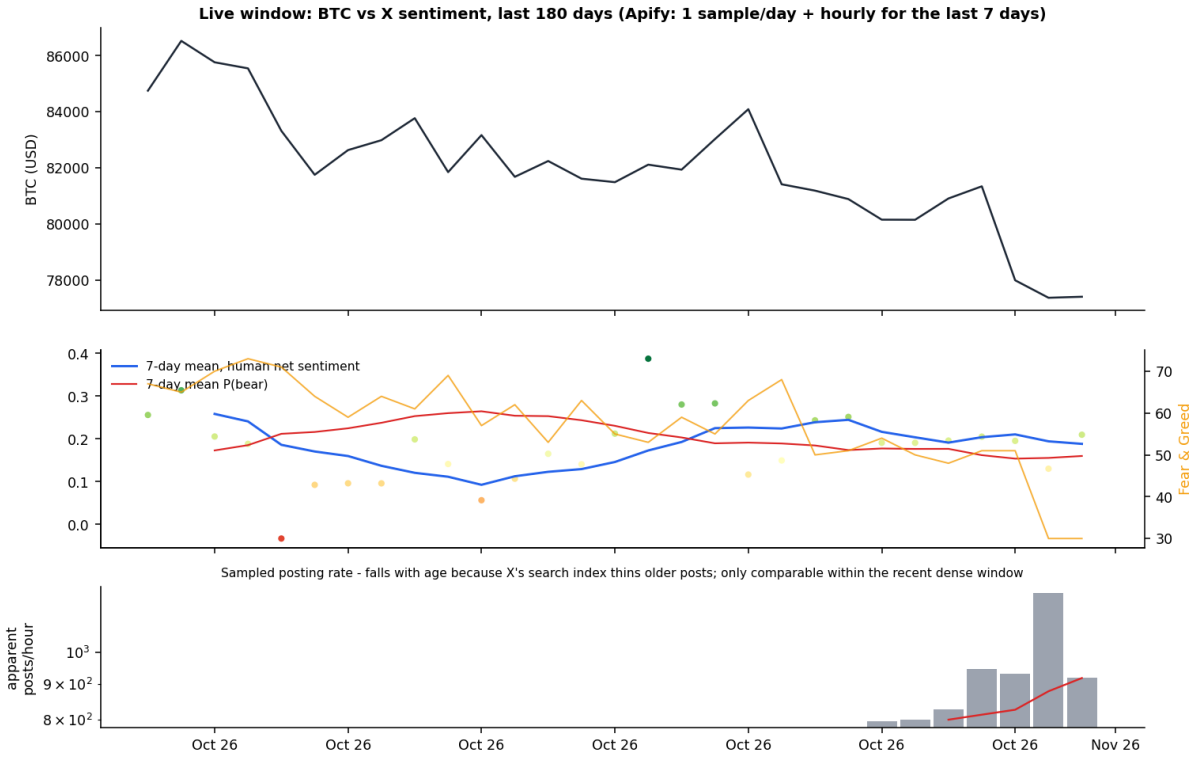


Feature importance (gain).

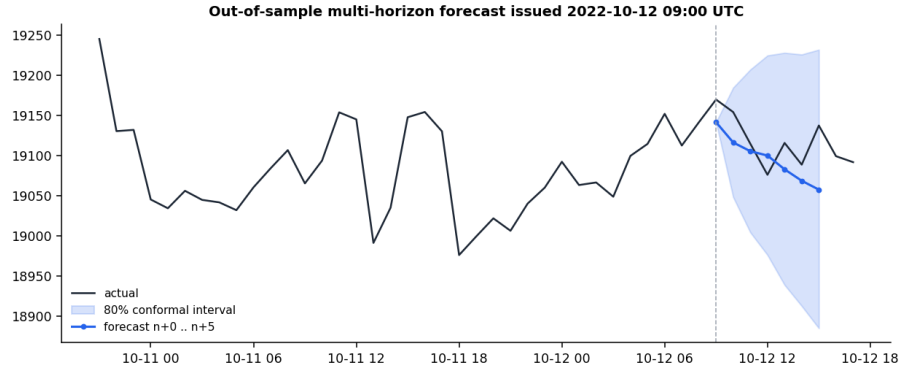
8. Live window (Apify, last 180 days)

Model (trained on 2021-23 only)	h	OOS R^2 vs RW	Dir. acc.	PI80 cover
rw	n+0	+0.0000	0.000	0.79
rw	n+5	+0.0000	0.000	0.77
price_lgbm	n+0	+0.0182	0.583	0.78
price_lgbm	n+5	-0.0501	0.442	0.74
full_lgbm	n+0	+0.0112	0.545	0.79
full_lgbm	n+5	-0.0494	0.513	0.74

156 dense hourly samples, 2026-10-25 06:00 → 2026-10-31 17:00 UTC. Too few hours for inference — a transfer smoke test showing the 2021-23 model and its intervals behave sensibly in 2026.



Live 180-day window: price, X sentiment, attention, Fear & Greed.



Example out-of-sample multi-horizon forecast with 80% conformal band.

9. Self-critique and threats to validity

- Historical X data is a sampled, gappy corpus (2021-23); live data is a budget sample (~20 posts per sampled hour). Neither is the full firehose. Effects that need dense data (minute-level, Guégan & Renault 2021) are out of reach.
- The test period is dominated by the 2022 bear market. Results may differ in trending bull regimes.
- Low-turnover and vol-managed variants were added after seeing round-1 results. They are disclosed and counted in the DSR rather than hidden. The volatility hypothesis was pre-specified from the literature, not data-mined.
- The sentiment model is trained on StockTwits labels and applied to X (domain shift). A CryptoBERT re-score of a sample is the next check.
- Post-level bot detection is heuristic (no ground truth). The all-posts vs human ablation shows no significant difference in accuracy.
- With a paid Apify plan the same code scales to dense hourly coverage for 180 days (≈ 430k posts, ≈ US$110–170). That is the single biggest improvement available.

10. Reproduce

pip install -r requirements.txt; cp .env.example .env (add APIFY_TOKEN, TAVILY_API_KEY); then make all (audit → data → sentiment → hourly → backtest → vol study → charts → scrape → live → report). make test runs the look-ahead tests.